

A counterexample to the Hilbert-Schmidt block implication in arXiv:1106.5089

Lane 4 candidate packet for fa_banach.001

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Source question

The source paper is Castillo–Defant–García–Pérez–García–Suárez, *Local complementation and the extension of bilinear mappings*, arXiv:1106.5089 [1]. In the remarks preceding the c_0 -conjecture, the authors observe that if a subspace $Y \subset c_0$ admits extension of bilinear forms to c_0 , then Y is Hilbert-Schmidt: every operator $Y \rightarrow \ell_2$ is 2-summing. For a block subspace $Y = c_0(F_n)$, this gives a uniform constant C such that, for every n ,

$$\pi_2(T) \leq C\|T\|, \quad T : F_n \longrightarrow \ell_2^n.$$

They then write that they do not know whether this condition implies that the finite-dimensional spaces F_n are uniformly complemented in finite-dimensional ℓ_∞ spaces.

can be C -complemented in ℓ_∞^m for some m depending only on n .

Remarks. There are aspects of this problem not well understood. Observe that a subspace Y of c_0 is locally complemented if and only if it is an $\mathcal{L}_\infty$ -space, in which case [2] it must be isomorphic to c_0 . Thus, $c_0(F_n)$ is not locally complemented in c_0 unless it is isomorphic to c_0 ; which implies that the spaces F_n must be uniformly complemented in c_0 , hence in some ℓ_∞^m . It is however a hard open problem, see [34], to determine if a sequence F_n of finite dimensional Banach spaces uniformly complemented in ℓ_∞^m must be uniformly isomorphic to ℓ_∞^k . On the other hand, it follows from [12] that if a subspace Y of c_0 admits the extension of bilinear forms to c_0 then it is a Hilbert-Schmidt space, namely $\mathfrak{L}(Y, \ell_2) = \Pi_2(Y, \ell_2)$. For a subspace of the form $c_0(F_n)$ this condition implies that there is a uniform constant C so that for every n the π_2 -summing norm and the standard operator norm are C -equivalent on $\mathfrak{L}(F_n, \ell_2^n)$. We do not know if this condition implies that the F_n are uniformly complemented in ℓ_∞^m . All this together suggests:

The c_0 -conjecture. A subspace of c_0 on which every bilinear form can be extended to the whole c_0 is isomorphic to c_0 .

3.2. Bilinear Johnson-Zippin theorem. The Johnson-Zippin theorem [27] establishes that $\mathcal{L}_\infty$ -valued operators defined on weak*-subspaces of ℓ_1 can be extended to the whole ℓ_1 . Again, there is no bilinear Johnson-Zippin theorem. Indeed, one has

Proof intuition

The 2-summing condition only tests Hilbert-valued operators. For ℓ_1^n , such operators are controlled uniformly by Grothendieck's inequality: the Gram matrix of the images of the coordinate vectors is a positive semidefinite matrix with bounded diagonal, and Grothendieck converts the Hilbert-vector estimate into the weak ℓ_2 norm appearing in the definition of π_2 .

Uniform complemented copies inside ℓ_∞ , however, would force uniform finite-dimensional L_∞ structure. Equivalently, by injectivity of ℓ_∞^m , it would give uniformly bounded absolute projection constants for ℓ_1^n . Those constants grow like $\sqrt{n}$. Thus the Hilbert-Schmidt test and the L_∞ -complementability conclusion separate.

The counterexample

Theorem 1. *Let $F_n = \ell_1^n$. Then there is a universal constant C such that*

$$\pi_2(T) \leq C\|T\| \quad \text{for every } n \text{ and every } T : \ell_1^n \rightarrow \ell_2^n.$$

Nevertheless the spaces ℓ_1^n are not uniformly complemented in finite-dimensional ℓ_∞ spaces. Hence the finite-dimensional Hilbert-Schmidt condition in the source remarks does not imply uniform ℓ_∞ -complementability of the blocks.

Proof. It is enough to prove the 2-summing estimate for operators $T : \ell_1^n \rightarrow H$, where H is an arbitrary Hilbert space. Normalize $\|T\| \leq 1$, and write $h_i = Te_i$, so that $\|h_i\| \leq 1$.

We write the scalar computation in the real case. The complex case is identical after replacing x_j^r by $\overline{x_j^r}$ in the Gram matrix and signs by unimodular scalars. Let $x^1, \dots, x^N \in \ell_1^n$. Put

$$a_{ij} = \sum_{r=1}^N x_i^r x_j^r.$$

The matrix $A = (a_{ij})$ is positive semidefinite. By Grothendieck's inequality in its positive semidefinite form, there is a universal constant K_G such that, for all Hilbert vectors u_i of norm at most one,

$$\sum_{i,j=1}^n a_{ij} \langle u_i, u_j \rangle \leq K_G \sup_{\varepsilon_i = \pm 1} \sum_{i,j=1}^n a_{ij} \varepsilon_i \varepsilon_j.$$

Applying this to $u_i = h_i$ gives

$$\begin{aligned} \sum_{r=1}^N \|Tx^r\|^2 &= \sum_{i,j=1}^n a_{ij} \langle h_i, h_j \rangle \\ &\leq K_G \sup_{\varepsilon_i = \pm 1} \sum_{r=1}^N \left| \sum_{i=1}^n \varepsilon_i x_i^r \right|^2 \\ &\leq K_G \sup_{\varphi \in B(\ell_1^n)^*} \sum_{r=1}^N |\varphi(x^r)|^2. \end{aligned}$$

This is precisely the defining inequality for $\pi_2(T) \leq \sqrt{K_G}$. Since ℓ_2^n is a Hilbert space, the displayed estimate proves the required uniform Hilbert-Schmidt condition.

It remains to show that the blocks are not uniformly complemented in finite-dimensional ℓ_∞ spaces. Suppose, to the contrary, that there were constants D, K , embeddings

$$J_n : \ell_1^n \longrightarrow \ell_\infty^{m(n)}$$

with $\|J_n\| \|J_n^{-1}\| \leq D$, and projections

$$P_n : \ell_\infty^{m(n)} \longrightarrow J_n(\ell_1^n)$$

with $\|P_n\| \leq K$. Because finite-dimensional $\ell_\infty^{m(n)}$ is injective, any isometric copy of ℓ_1^n inside any Banach space would then be DK -complemented: extend the map J_n from that copy into $\ell_\infty^{m(n)}$, compose with P_n , and then with J_n^{-1} .

Thus the absolute projection constants of ℓ_1^n would be uniformly bounded. This contradicts the classical Grunbaum–Rutovitz estimate

$$\lambda(\ell_1^n) \asymp \sqrt{n},$$

where $\lambda(E)$ denotes the absolute projection constant of the finite-dimensional space E . Therefore no such uniform complemented ℓ_∞ -realization exists. $\square$

Limitations

This packet answers only the finite-dimensional implication asked in the remarks before the c_0 -conjecture. It does not construct a subspace $Y \subset c_0$ for which all bilinear forms extend to c_0 , and it does not disprove the c_0 -conjecture itself. It shows that the Hilbert-Schmidt necessary condition isolated in the source is too weak, by itself, to force the block complementability conclusion.

Verification and novelty notes

The proof uses only two standard external facts: Grothendieck’s inequality and the unbounded absolute projection constants of ℓ_1^n . A bounded novelty check on 2026-06-29 searched the run indexes for arXiv:1106.5089 and the keywords “Hilbert-Schmidt”, “ π_2 ”, “uniformly complemented”, and “ ℓ_1^n ”; it also read the two previous local attempts for this paper. Web searches for the exact source phrase and close variants did not find an explicit later answer. Since the counterexample is a short combination of classical facts, novelty confidence should be treated as modest: this may be a folklore observation even if no exact citation was found in the bounded search.

Recommended human-review focus: check that the source’s phrase “uniformly complemented in ℓ_∞^m ” is intended in the usual uniform isomorphic-complemented sense. Under that standard reading, the projection-constant argument above is decisive.

References

- [1] J. M. F. Castillo, A. Defant, R. Garcia, D. Perez-Garcia, and J. Suarez, *Local complementation and the extension of bilinear mappings*, arXiv:1106.5089, 2011.
- [2] A. Grothendieck, *Resume de la theorie metrique des produits tensoriels topologiques*, Bol. Soc. Mat. Sao Paulo **8** (1953), 1–79.
- [3] B. Grunbaum, *Projection constants*, Trans. Amer. Math. Soc. **95** (1960), 451–465.
- [4] D. Rutovitz, *Some parameters associated with finite-dimensional Banach spaces*, J. London Math. Soc. **40** (1965), 241–255.